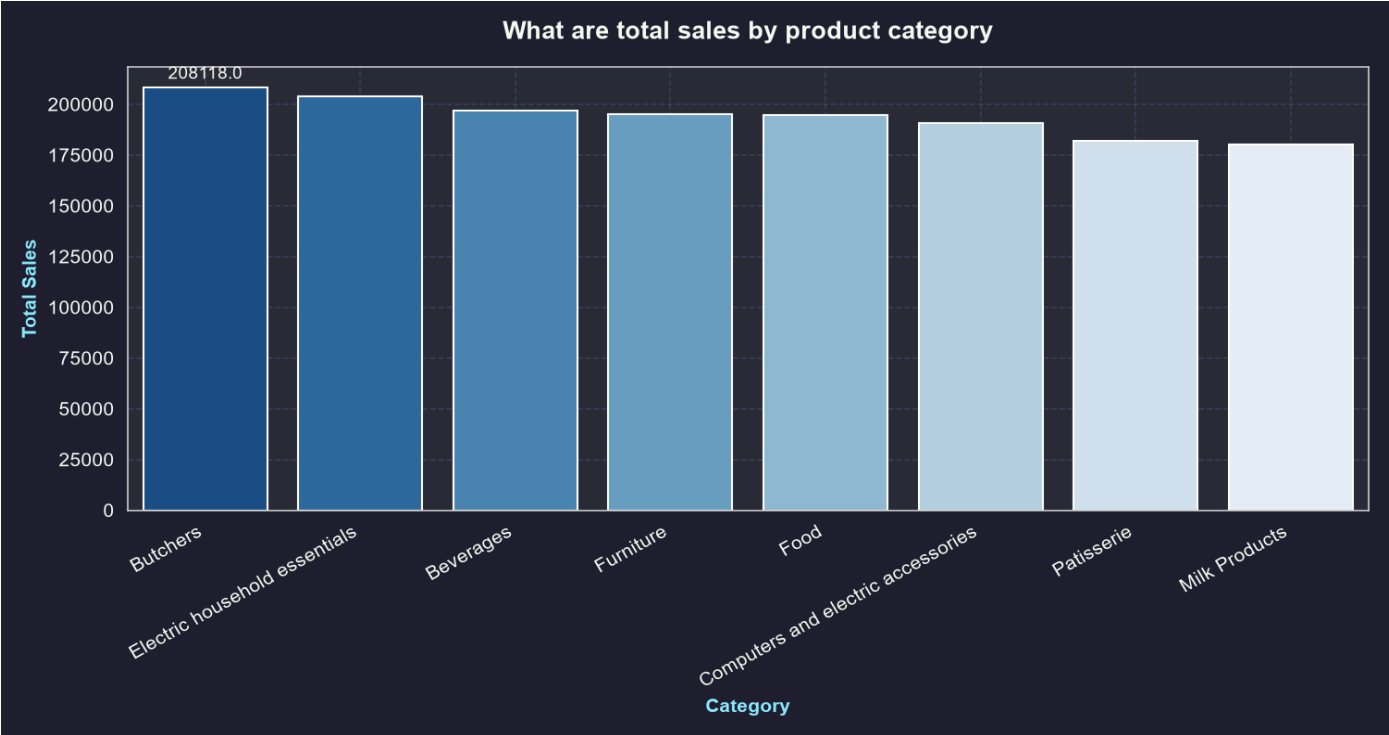


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** What are total sales by product category?
- **Answer:** The aggregate sales (in USD) for each product category are:

Rank	Category	Total Sales (USD)
1	Butchers	208,118.00
2	Electric household essentials	203,813.50
3	Beverages	197,047.50
4	Furniture	195,310.00
5	Food	194,812.00
6	Computers and electric accessories	190,692.50
7	Patisserie	182,165.50

| 8 | Milk Products | **180,112.00** |

The list is sorted from highest to lowest total sales, revealing that **Butchers** leads the portfolio, followed closely by **Electric household essentials** and **Beverages**.

2. Dataset Overview & Data Quality Profile

| Attribute | Detail |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Primary Identifier** | Not explicitly defined (transaction-level rows) |

| **Missing Values** | Detected in: **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, **Discount Applied** |

| **Duplicates** | None (0) |

| **Potential Date Columns** | **Transaction Date** stored as string |

| **Data Quality Issues** | Missing values in the columns listed above; date column may need parsing for time-series work. |

3. Data Cleaning & Validation Summary

• **Cleaning Action:** Removed duplicate rows (none found) and created a clean copy.

```
result = df.drop_duplicates().copy()
```

• **Validation Outcome:** All 12,575 rows retained; dataset considered *validated* for subsequent analysis.

4. User Query & Analytical Approach

• **User Question:** “What are total sales by product category?”

• **Analytical Methodology:**

1. **Group** the data by the **Category** column.
2. **Aggregate** the **Total Spent** values using a sum to compute total sales per category.
3. **Rename** the aggregated column to **Total Sales** for clarity.
4. **Sort** the result in descending order of sales for easy interpretation.

5. Python Analysis Code & Executed Data Output

Code

```
# Aggregate total sales per product category result = ( df.groupby('Category',
as_index=False)['Total Spent'] .sum() .rename(columns={'Total Spent': 'Total Sales'}) ) # Sort
categories by descending sales result = result.sort_values('Total Sales',
ascending=False).reset_index(drop=True)
```

Executed Output (first 8 rows)

Category	Total Sales
Butchers	208,118.0
Electric household essentials	203,813.5
Beverages	197,047.5
Furniture	195,310.0
Food	194,812.0
Computers and electric accessories	190,692.5
Patisserie	182,165.5
Milk Products	180,112.0

(The full result DataFrame contains one row per distinct product category.)

6. Key Insights & Strategic Recommendations

Insights

- **Top■Performing Category:** *Butchers* generates the highest revenue, indicating strong demand or higher price points.
- **Close Competitors:** *Electric household essentials* and *Beverages* are within ~5% of the leader, suggesting they are also strategic pillars.
- **Mid■Tier Segments:** *Furniture*, *Food*, and *Computers & electric accessories* each contribute roughly \$190■\$195 k, representing a solid revenue base.
- **Lower■Tier Segments:** *Patisserie* and *Milk Products* are the smallest contributors among the listed categories but still exceed \$180 k.

Recommendations

1. Invest in High■Margin Categories

- Conduct a deeper profitability analysis (e.g., gross margin) for *Butchers* and *Electric household essentials* to confirm that high sales translate to high profit.
- Allocate marketing budget proportionally to sustain or grow these segments.

2. Cross-Sell Opportunities

- Bundle *Beverages* with *Food* or *Patisserie* in promotional packages to lift the latter's sales.

3. Category Expansion Assessment

- Evaluate whether the *Milk Products* and *Patisserie* categories have growth potential (e.g., new SKUs, seasonal promotions) to close the gap with higher-performing groups.

4. Data Enrichment

- Parse **Transaction Date** into a proper datetime type to enable time-series trend analysis (seasonality, growth rates).
- Address missing values in **Item**, **Price Per Unit**, **Quantity**, **Total Spent**, and **Discount Applied** to improve the reliability of downstream metrics such as average transaction value or discount effectiveness.

5. Continuous Monitoring

- Set up a monthly dashboard that tracks **Total Sales** by category, flagging any category that drops >10% month-over-month for rapid investigation.

Prepared by the Autonomous AI Data Analyst

![What are total sales by product category? Chart](analysis_chart.png)